

1. Introduction

This report presents a high-level Data Warehouse (DWH) design for a movie rental business. The transition from an OLTP schema to a dimensional model aims to support complex analytical queries that the original normalized database cannot handle efficiently. By using a star schema, the design improves query performance and provides a consolidated view of business operations for decision-makers. The OLTP schema is used as a source to understand the data structure, while the data warehouse is a redesigned model optimized for analytical processing.

2. Business Questions

2.1 Rental Activity Analysis (Fact_Rental)

- Which films are rented most frequently?
Importance: Identifying trending content helps optimize the catalog and ensure availability of high-demand movies.
- Which film categories are most popular?
Importance: Supports strategic decisions on investing in genres that drive engagement.
- What is the average rental duration for different films or categories?
Importance: Provides insights into customer viewing behavior and helps adjust rental policies.
- Which films are returned late most often?
Importance: Helps identify inventory bottlenecks and improve late return policies.

2.2 Revenue Analysis (Fact_Payment)

- Which films generate the highest revenue?
Importance: Identifies the most profitable titles.
- How does revenue change over time (monthly, quarterly, yearly)?
Importance: Supports trend analysis and forecasting.
- Which customers generate the highest revenue?
Importance: Helps identify high-value customers.

2.3 Store and Geographic Performance (Fact_Inventory & Fact_Rental)

- Which stores generate the highest revenue?
Importance: Evaluates store performance and supports expansion decisions.

- Which cities or countries generate the highest activity?
Importance: Helps target marketing efforts geographically.

3. Dimensional Model Design

3.1 Selected Business Processes

- Rental Events
- Payment Transactions
- Inventory Snapshots

3.2 Fact Tables

Fact_Rental

- Business Process: Movie rental transactions
- Grain: One row per rental transaction
- Measures:
 - Rental_Count
 - Rental_Duration
 - Late_Return_Flag
- Related Dimensions:
 - Dim_Date, Dim_Customer, Dim_Film, Dim_Store

Fact_Payment

- Business Process: Payment transactions
- Grain: One row per payment
- Measures:
 - Payment_Amount
- Related Dimensions:
 - Dim_Date, Dim_Customer, Dim_Staff

Fact_Inventory_Daily_Snapshot

- Business Process: Daily inventory tracking
- Grain: One row per film, per store, per day
- Measures:
 - In_Stock_Quantity
 - Rented_Out_Quantity
- Related Dimensions:
 - Dim_Date, Dim_Film, Dim_Store

Note:

This is a snapshot fact table that captures the state of inventory at a specific point in time rather than recording transactional events.

3.3 Dimension Tables

Dim_Film

- Attributes: Film_Title, Description, Release_Year, Rental_Duration_Policy, Rental_Rate, Length, Rating, Language_Name, Category_Name
- Source: film, language, category, film_category

Dim_Customer

- Attributes: Full_Name, Email, Active_Status, Address, District, City, Country
- Source: customer, address, city, country

Dim_Store

- Attributes: Store_ID, Manager_Full_Name, Store_Address, Store_City, Store_Country
- Source: store, staff, address, city, country

Dim_Staff

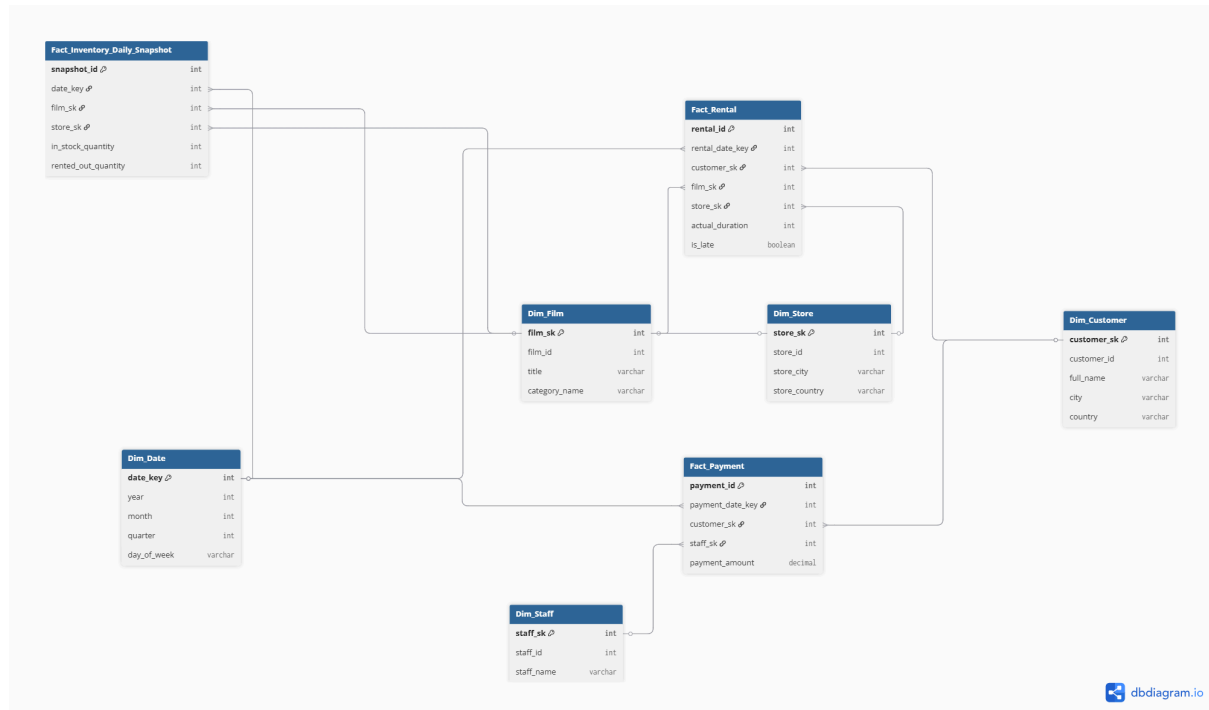
- Attributes: Staff_Full_Name, Username, Email, Active_Status, Store_ID
- Source: staff

Dim_Date

- Attributes: Date_Key, Full_Date, Day_of_Week, Month, Quarter, Year, Is_Weekend, Is_Holiday
- The Date_Key is a surrogate key in the format YYYYMMDD.

4. Dimensional Model Diagram

The model follows a star schema where fact tables are connected to shared conformed dimensions such as Dim_Date and Dim_Film.



5. ETL Design

5.1 Extract

Data is extracted from OLTP tables such as rental, payment, inventory, film, and customer.

5.2 Transform

- Join related tables to create dimension tables
- Convert date fields into proper formats
- Calculate rental duration
- Merge rental and film data to obtain correct rental duration policies
- Detect late returns
- Handle missing values
- Generate surrogate keys

Sample Code

```
import pandas as pd
import numpy as np

df_rental = pd.read_csv('rental.csv')
df_film = pd.read_csv('film.csv')
df_category = pd.read_csv('category.csv')
df_film_cat = pd.read_csv('film_category.csv')

df_merged_film = pd.merge(df_film, df_film_cat, on='film_id', how='left')
dim_film = pd.merge(df_merged_film, df_category, on='category_id', how='left')

df_rental['rental_date'] = pd.to_datetime(df_rental['rental_date'])
df_rental['return_date'] = pd.to_datetime(df_rental['return_date'])

df_rental['actual_duration'] = (df_rental['return_date'] - df_rental['rental_date']).dt.days

df_rental = pd.merge(df_rental, df_film[['film_id', 'rental_duration']], on='film_id', how='left')

df_rental['is_late'] = df_rental['actual_duration'] > df_rental['rental_duration']
df_rental['is_late'] = df_rental['is_late'].fillna(False)

dim_film['film_sk'] = np.arange(len(dim_film)) + 1
```

5.3 Load

- Dimension tables are loaded first
- Fact tables are loaded afterward

During the loading of Fact_Rental, a lookup is performed on Dim_Film using the operational film_id to replace it with the generated surrogate key (film_sk), ensuring proper relationships in the data warehouse.

6. Data Quality

- No null values in key fields
- Payment amounts must be positive
- Return dates must be valid
- Referential integrity must be maintained

7. Conclusion

The dimensional model transforms the OLTP system into a structure suitable for analytical processing. It enables better insights into customer behavior, revenue, and inventory, supporting more effective decision-making.